

MRC Lecture 1 — Derivation Guide 5

The Temperature Parameter: Controlling the Sharpness of Selection

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1. Motivation

In Derivation Guide 4, we defined the softmax function and showed that it converts scores into probabilities. But the standard softmax treats all score differences equally — a difference of 2 always produces the same probability ratio ($\exp(2) \approx 7.4$), regardless of the context.

In physical systems, the sharpness of selection is tunable. At very low temperatures, a chemical system locks into its lowest-energy state — selection is nearly deterministic. At very high temperatures, thermal fluctuations overwhelm energy differences and all states are occupied nearly equally — selection is nearly random. At physiological temperature (37°C), the thermal energy $k_B T \approx 0.6$ kcal/mol is comparable to the energy differences between molecular binding partners, producing **probabilistic selection** — the regime relevant to all biological molecular recognition. The **temperature parameter** τ introduces exactly this control into softmax. It is the single parameter that governs the stringency of selection, and it turns out to be the same parameter in both the Boltzmann distribution and transformer attention.

2. Definition: Temperature-Scaled Softmax

Definition. Given a score vector $z = (z_1, \dots, z_M)$ and a temperature parameter $\tau > 0$, the temperature-scaled softmax is:

$$P_j = \frac{\exp(z_j / \tau)}{\sum_{k=1}^M \exp(z_k / \tau)}$$

The only change from the standard softmax (Derivation Guide 4) is that every score is divided by τ before exponentiation.

Equivalence to scaling. Dividing scores by τ is equivalent to multiplying scores by $1/\tau$. From Checkpoint Question Q4 of Derivation Guide 4, we know that scaling scores changes the output distribution. Temperature is the parameter that controls this scaling.

3. Three Regimes of Temperature

The temperature parameter creates three qualitatively distinct selection regimes.

Low temperature ($\tau \rightarrow 0^+$): sharp selection.

As τ decreases, z_j / τ becomes very large for the highest score and very negative for lower scores. The exponential amplifies these differences, concentrating nearly all probability on the single highest-scoring candidate:

$$\lim_{\tau \rightarrow 0^+} \text{softmax}(z / \tau) = \text{argmax}(z)$$

In this limit, softmax becomes argmax — deterministic, winner-take-all selection. Biologically, this corresponds to a highly specific molecular interaction where only the optimal binding partner is selected.

High temperature ($\tau \rightarrow \infty$): flat selection.

As τ increases, z_j / τ approaches zero for all j , and $\exp(z_j / \tau)$ approaches $\exp(0) = 1$ for all j . The probability distribution approaches uniform:

$$\lim_{\tau \rightarrow \infty} \text{softmax}(z/\tau) = \left(\frac{1}{M}, \frac{1}{M}, \dots, \frac{1}{M} \right)$$

All candidates receive equal probability regardless of their scores. Biologically, this corresponds to non-specific binding where the selection agent has no preference among targets.

Intermediate temperature: probabilistic selection.

At intermediate τ , the probability distribution is graded — higher scores receive higher probability, but lower-scoring candidates retain non-negligible probability. This is the physically relevant regime for biological molecular recognition at physiological temperature.

4. Worked Example: Three Temperatures

Problem. Compute $\text{softmax}(z/\tau)$ for $z = (2.0, 1.0, 0.1)$ at three temperatures: $\tau = 0.1$, $\tau = 1.0$, and $\tau = 10.0$.

Case 1: $\tau = 0.1$ (low temperature — sharp)

Step 1. Divide scores by τ :

$$z/\tau = (2.0/0.1, 1.0/0.1, 0.1/0.1) = (20, 10, 1)$$

Step 2. Exponentiate:

$$\exp(20) = 4.851 \times 10^8, \exp(10) = 2.203 \times 10^4, \exp(1) = 2.718$$

Step 3. Partition function:

$$Z = 4.851 \times 10^8 + 2.203 \times 10^4 + 2.718 \approx 4.851 \times 10^8$$

The first term dominates overwhelmingly.

Step 4. Probabilities:

$$P_1 = \frac{4.851 \times 10^8}{4.851 \times 10^8} \approx 1.000$$

$$P_2 = \frac{2.203 \times 10^4}{4.851 \times 10^8} \approx 0.000045$$

$$P_3 = \frac{2.718}{4.851 \times 10^8} \approx 0.0000000056$$

Result: $P \approx (1.000, 0.000, 0.000)$. Nearly all probability is on the highest score. This is approaching argmax .

Case 2: $\tau = 1.0$ (standard temperature)

This is the same computation as Worked Example 1 in Derivation Guide 4.

Step 1. $z/\tau = (2.0, 1.0, 0.1)$ (unchanged).

Step 2. $\exp(2.0) = 7.389$, $\exp(1.0) = 2.718$, $\exp(0.1) = 1.105$.

Step 3. $Z = 11.212$.

Step 4. Probabilities:

$$P_1 = 0.659, P_2 = 0.242, P_3 = 0.099$$

Result: $P = (0.659, 0.242, 0.099)$. The ranking is preserved, but all candidates retain meaningful probability. This is probabilistic selection.

Case 3: $\tau = 10.0$ (high temperature — flat)

Step 1. Divide scores by τ :

$$z/\tau = (2.0/10, 1.0/10, 0.1/10) = (0.2, 0.1, 0.01)$$

Step 2. Exponentiate:

$$\exp(0.2) = 1.221, \exp(0.1) = 1.105, \exp(0.01) = 1.010$$

Step 3. Partition function:

$$Z = 1.221 + 1.105 + 1.010 = 3.336$$

Step 4. Probabilities:

$$P_1 = \frac{1.221}{3.336} = 0.366$$

$$P_2 = \frac{1.105}{3.336} = 0.331$$

$$P_3 = \frac{1.010}{3.336} = 0.303$$

Result: $P = (0.366, 0.331, 0.303)$. The distribution is nearly uniform. The score differences (2.0 vs. 1.0 vs. 0.1) are almost completely washed out by the high temperature. This is approaching $1/M = 1/3 = 0.333$ for all candidates.

Summary of the Three Cases

	$z_1 = 2.0$	$z_2 = 1.0$	$z_3 = 0.1$	Character
$\tau = 0.1$	1.000	0.000	0.000	Nearly deterministic
$\tau = 1.0$	0.659	0.242	0.099	Probabilistic
$\tau = 10.0$	0.366	0.331	0.303	Nearly uniform

The same score vector produces three qualitatively different selection outcomes depending on the temperature. Temperature is the dial that tunes selection stringency.

5. Temperature and the Probability Ratio

The temperature parameter controls how strongly score differences translate into probability differences.

Claim. For any two candidates A and B with score difference $\delta = z_A - z_B$:

$$\frac{P_A}{P_B} = \exp\left(\frac{\delta}{\tau}\right)$$

Proof. From the IIA property (Derivation Guide 4, §5):

$$\frac{P_A}{P_B} = \frac{\exp(z_A/\tau)}{\exp(z_B/\tau)} = \exp\left(\frac{z_A - z_B}{\tau}\right) = \exp\left(\frac{\delta}{\tau}\right) \blacksquare$$

Interpretation. The effective score difference is δ/τ . At low temperature, even a small δ produces a large ratio. At high temperature, even a large δ produces a ratio close to 1.

Numerical examples for $\delta = 1.0$:

Temperature τ	Effective difference δ/τ	Probability ratio $\exp(\delta/\tau)$
0.1	10	22,026
0.5	2	7.39
1.0	1	2.72
2.0	0.5	1.65
10.0	0.1	1.11

At $\tau = 0.1$, a unit score difference produces a 22,000-fold probability advantage. At $\tau = 10$, the same difference produces only an 11% advantage. Temperature determines the “resolution” of the selection process — how sensitively it discriminates between candidates with similar scores.

6. Temperature in Molecular Binding: $\tau = k_B T$

In the Boltzmann distribution governing molecular binding at thermal equilibrium, the temperature parameter has a precise physical identity.

The Boltzmann distribution (Lecture 4). The probability of an agent binding target j with binding free energy ΔG_j is:

$$P(j) = \frac{\exp(-\Delta G_j / k_B T)}{\sum_k \exp(-\Delta G_k / k_B T)}$$

where $k_B = 1.987 \times 10^{-3}$ kcal/(mol · K) is Boltzmann's constant and T is the absolute temperature in Kelvin.

Identification with softmax. Define the dimensionless score $s_j = -\Delta G_j / k_B T$. Then:

$$P(j) = \frac{\exp(s_j)}{\sum_k \exp(s_k)} = \text{softmax}(s)_j$$

This is the standard softmax with $\tau = 1$ in natural units. Equivalently, working in energy units with $s_j = -\Delta G_j$:

$$P(j) = \frac{\exp(-\Delta G_j / k_B T)}{\sum_k \exp(-\Delta G_k / k_B T)} = \text{softmax}(-\Delta G / \tau)_j$$

with $\tau = k_B T$.

Physiological values. At 37°C ($T = 310$ K):

$$k_B T = 1.987 \times 10^{-3} \times 310 = 0.616 \text{ kcal/mol}$$

This is the thermal energy available per molecule. It sets the resolution of molecular selection: binding energy differences smaller than ~ 0.6 kcal/mol are partially “washed out” by thermal fluctuations, while differences much larger than 0.6 kcal/mol produce strong selectivity.

Worked example. An antibody encounters two antigens. Antigen A has $\Delta G_A = -10.0$ kcal/mol (tight binder). Antigen B has $\Delta G_B = -8.0$ kcal/mol (moderate binder). The energy difference is 2.0 kcal/mol.

At $T = 310$ K:

$$\frac{P_A}{P_B} = \exp\left(\frac{-\Delta G_A - (-\Delta G_B)}{k_B T}\right) = \exp\left(\frac{10.0 - 8.0}{0.616}\right) = \exp(3.25) \approx 25.8$$

The antibody is about 26 times more likely to bind Antigen A than Antigen B. The 2 kcal/mol energy difference, amplified by the Boltzmann exponential at physiological temperature, produces strong but not absolute selectivity — Antigen B retains a non-zero binding probability.

7. Temperature in Transformer Attention: $\tau = \sqrt{d_k}$

In the transformer attention mechanism (Vaswani et al., 2017), the temperature parameter has a different origin but plays the same mathematical role.

Scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

The scaling factor $\sqrt{d_k}$ serves as the temperature: $\tau = \sqrt{d_k}$.

Why $\sqrt{d_k}$? Vaswani et al. provided a statistical argument. Assume the components of q and k are

independent random variables with mean 0 and variance 1. Then the dot product $q \cdot k = \sum_{l=1}^{d_k} q_l k_l$ is a

sum of d_k independent random variables, each with mean 0 and variance 1.

By the properties of sums of independent random variables:

$$E[q \cdot k] = \sum_{l=1}^{d_k} E[q_l k_l] = \sum_{l=1}^{d_k} E[q_l] E[k_l] = 0$$

$$\text{Var}(q \cdot k) = \sum_{l=1}^{d_k} \text{Var}(q_l k_l) = \sum_{l=1}^{d_k} 1 = d_k$$

The variance of the dot product grows linearly with the key dimension d_k . Without scaling, as d_k increases (e.g., $d_k = 512$), the dot products become large, softmax saturates toward a one-hot vector, and gradients vanish. Dividing by $\sqrt{d_k}$ restores unit variance:

$$\text{Var}\left(\frac{q \cdot k}{\sqrt{d_k}}\right) = \frac{d_k}{d_k} = 1$$

The connection. Vaswani's team chose $\sqrt{d_k}$ for numerical stability. The Boltzmann distribution uses $k_B T$ because that is the physical temperature. Both serve the same purpose: controlling the sharpness of the probability distribution produced by softmax. In SFMs, τ is a **learned parameter** (following CLIP), and the prediction is that τ_{learned} should recover a value proportional to $k_B T$, up to a unit conversion determined by the embedding geometry. This prediction is testable through the affinity calibration framework (Lecture 10).

8. Worked Example: Temperature and Molecular Selectivity

Problem. A transcription factor encounters three candidate DNA binding sites with the following binding free energies (more negative = tighter binding):

$$\Delta G_1 = -8.0 \text{ kcal/mol}, \Delta G_2 = -7.0 \text{ kcal/mol}, \Delta G_3 = -5.0 \text{ kcal/mol}$$

Compute the binding probability distribution at three temperatures: $T = 155 \text{ K}$ (half physiological), $T = 310 \text{ K}$ (physiological), and $T = 620 \text{ K}$ (double physiological).

Setup. The scores are $s_j = -\Delta G_j / k_B T$, and the temperature parameter is $\tau = k_B T$.

At $T = 155 \text{ K}$ ($k_B T = 0.308 \text{ kcal/mol}$):

$$s_1 = 8.0 / 0.308 = 26.0, s_2 = 7.0 / 0.308 = 22.7, s_3 = 5.0 / 0.308 = 16.2$$

The score differences are very large. Softmax at these values gives $P_1 \approx 0.965$, $P_2 \approx 0.035$, $P_3 \approx 0.000$. The tightest binder dominates almost completely.

At $T = 310 \text{ K}$ ($k_B T = 0.616 \text{ kcal/mol}$):

$$s_1 = 8.0 / 0.616 = 13.0, s_2 = 7.0 / 0.616 = 11.4, s_3 = 5.0 / 0.616 = 8.1$$

Using the shift trick (subtract max score 13.0): $\tilde{s} = (0, -1.6, -4.9)$.

$$\exp(0) = 1.000, \exp(-1.6) = 0.202, \exp(-4.9) = 0.0074$$

$$Z = 1.000 + 0.202 + 0.0074 = 1.209$$

$$P_1 = 0.827, P_2 = 0.167, P_3 = 0.006$$

The tightest binder is strongly preferred but Site 2 retains meaningful probability.

At $T = 620 \text{ K}$ ($k_B T = 1.232 \text{ kcal/mol}$):

$$s_1 = 8.0 / 1.232 = 6.49, s_2 = 7.0 / 1.232 = 5.68, s_3 = 5.0 / 1.232 = 4.06$$

Shift: $\tilde{s} = (0, -0.81, -2.43)$.

$$\exp(0) = 1.000, \exp(-0.81) = 0.445, \exp(-2.43) = 0.088$$

$$Z = 1.000 + 0.445 + 0.088 = 1.533$$

$$P_1 = 0.652, P_2 = 0.290, P_3 = 0.057$$

Selection is more diffuse. All three sites have non-negligible probability.

Summary:

Temperature	$k_B T$ (kcal/mol)	P_1 ($\Delta G = -8.0$)	P_2 ($\Delta G = -7.0$)	P_3 ($\Delta G = -5.0$)
$T = 155 \text{ K}$	0.308	0.965	0.035	0.000

Temperature	$k_B T$ (kcal/mol)	$P_1 (\Delta G = -8.0)$	$P_2 (\Delta G = -7.0)$	$P_3 (\Delta G = -5.0)$
$T = 310$ K	0.616	0.827	0.167	0.006
$T = 620$ K	1.232	0.652	0.290	0.057

As temperature increases, the probability distribution flattens. As temperature decreases, it sharpens. The same binding energies produce qualitatively different selection behavior depending on the thermal environment.

9. The Identity Preview

We can now state the identity that motivates this entire course, using all four operations from Derivation Guides 1–5.

Molecular binding (Boltzmann, 1877):

$$P(\text{bind } j) = \frac{\exp(-\Delta G_j / k_B T)}{\sum_k \exp(-\Delta G_k / k_B T)}$$

Transformer attention (Vaswani, 2017):

$$A_j = \frac{\exp(q \cdot k_j / \tau)}{\sum_k \exp(q \cdot k_k / \tau)}$$

Score identification:

$$s_j = -\Delta G_j / k_B T = q \cdot k_j / \tau$$

These are the same equation. Every component maps:

Molecular binding	Transformer attention	Role
Binding free energy $-\Delta G_j$	Dot product $q \cdot k_j$	Compatibility score
Thermal energy $k_B T$	Temperature τ (or $\sqrt{d_k}$)	Selection stringency
Partition function $\sum \exp(\cdot)$	Attention normalizer $\sum \exp(\cdot)$	Ensures probabilities sum to 1
Binding probability $P(j)$	Attention weight A_j	Selection probability

The dot product (Derivation Guide 1) computes the score. Matrix multiplication (Derivation Guide 2) computes all pairwise scores. Cosine similarity (Derivation Guide 3) normalizes the magnitude.

Softmax with temperature (Derivation Guides 4 and 5) converts scores to probabilities.

This identity — and the proof that it is unique (Lecture 6) — is the foundation of the entire course.

10. Temperature in SFMs: A Learned Parameter

In Specificity Foundation Models, the temperature τ is neither fixed by physics ($k_B T$) nor by heuristic ($\sqrt{d_k}$). Following the approach of CLIP (Radford et al., 2021), SFMs treat τ as a **learned parameter** that is optimized during contrastive training alongside the encoder weights.

Why learn temperature? The embedding space has an arbitrary scale — the magnitude of the dot product $q \cdot k$ depends on the projection head dimension, L2 normalization, and the internal geometry of the encoder. The learned τ adapts the softmax sharpness to the actual scale of the embedding-space scores.

The temperature recovery prediction. If the SFM architecture truly implements the Boltzmann distribution (Level 3 alignment), then the learned temperature should be related to the physical temperature. Specifically, the affinity calibration framework (Lecture 10) predicts that:

$$\frac{w_B}{w_A} = \tau_{\text{learned}}$$

where w_A is the slope of the affinity regression using post- τ scores and w_B is the slope using pre- τ raw dot products. If this ratio equals τ_{learned} , the learned temperature has the physical meaning assigned by the convergence equation. This is a testable, quantitative prediction.

Biological interpretation. The learned τ can be interpreted as the effective selection stringency of the molecular recognition system. A system with low learned τ (e.g., CRISPR on-target recognition) operates in a high-specificity regime. A system with high learned τ (e.g., innate immune receptors with broad cross-reactivity) operates in a low-specificity regime. Comparing learned temperatures across SFM domains tests whether the computational model recovers the known biological stringency hierarchy.

11. Summary

Concept	Definition	Significance
Temperature-scaled softmax	$P_j = \exp(z_j / \tau) / \sum_k \exp(z_k / \tau)$	One parameter controls selection sharpness
Low τ	$\tau \rightarrow 0^+$	Sharp selection $\rightarrow$ argmax (deterministic)
High τ	$\tau \rightarrow \infty$	Flat selection $\rightarrow$ uniform (no preference)
Intermediate τ	Physiological regime	Probabilistic selection
Probability ratio	$P_A / P_B = \exp(\delta / \tau)$	Score difference amplified by $1 / \tau$
Boltzmann: $\tau = k_B T$	0.616 kcal/mol at 310 K	Physical thermal energy
Transformer: $\tau = \sqrt{d_k}$	Variance normalization	Prevents softmax saturation
SFM: τ learned	Optimized during training	Testable via temperature recovery

12. Checkpoint Questions

Q1. At physiological temperature ($k_B T = 0.616$ kcal/mol), what probability ratio does a 1 kcal/mol binding energy difference produce? What about a 3 kcal/mol difference?

Answer: For $\delta = 1.0$ kcal/mol: $P_A / P_B = \exp(1.0 / 0.616) = \exp(1.623) \approx 5.1$. The tighter binder is about $5\times$ more likely. For $\delta = 3.0$ kcal/mol: $P_A / P_B = \exp(3.0 / 0.616) = \exp(4.87) \approx 131$. The tighter binder is about $130\times$ more likely. A 3 kcal/mol difference — roughly the difference between a nanomolar and a micromolar binder — produces overwhelming selectivity at physiological temperature.

Q2. A transformer uses key dimension $d_k = 64$. What is the temperature τ ? If the key dimension is increased to $d_k = 512$, how does this affect the attention distribution (sharper or flatter)?

Answer: At $d_k = 64$: $\tau = \sqrt{64} = 8$. At $d_k = 512$: $\tau = \sqrt{512} \approx 22.6$. Higher d_k means higher τ , which flattens the attention distribution. Without the $\sqrt{d_k}$ scaling, the raw dot products would be larger at higher dimensions (variance $= d_k$), causing softmax to saturate. The scaling compensates for this, but the effective temperature still increases with d_k .

Q3. Two SFMs are trained on different molecular recognition domains. The first (crisprSFM) learns $\tau = 0.05$. The second (rlSFM, for receptor-ligand) learns $\tau = 0.20$. What does this suggest about the relative stringency of molecular recognition in these two systems?

Answer: Lower τ means sharper selection — the system is more discriminating. The crisprSFM's low τ is consistent with CRISPR's requirement for high on-target specificity (a guide RNA must select one genomic site out of billions). The rlSFM's higher τ is consistent with receptor-ligand signaling, where receptors often bind multiple ligands with different affinities (cross-reactivity is a feature, not a bug). If these learned temperatures correlate with the known biological stringency of each system, it provides evidence that the SFM has captured the physical selection regime.

Q4. A student claims: "Temperature is just a hyperparameter — it's the same as the learning rate or batch size, just something you tune." How would you respond based on the MRC framework?

Answer: In standard machine learning, temperature is indeed a hyperparameter with no physical meaning. In the MRC framework, temperature is fundamentally different: it corresponds to the physical thermal energy $k_B T$ through the Boltzmann-softmax identity. The learning rate and batch size have no physical analogue — they are optimization parameters. Temperature is a physics

parameter that appears in the governing equation. The temperature recovery prediction (Lecture 10) provides a concrete test: if $w_B / w_A = \tau_{\text{learned}}$, the learned temperature has absorbed the physical temperature. No other hyperparameter makes a testable prediction connecting the trained model to a physical constant.

13. Connection to Future Lectures

- **Lecture 2:** The transformer attention equation uses $\tau = \sqrt{d_k}$. The variance argument for this scaling is derived on the whiteboard. Multi-head attention uses independent temperature-scaled attention in each head.
- **Lecture 4:** The Boltzmann distribution is derived from maximum entropy with $\tau = k_B T$. The physical meaning of temperature as thermal energy is formalized.
- **Lecture 6:** The convergence equation $P(j) = \exp(s_j / \tau) / \sum_k \exp(s_k / \tau)$ is proven to be unique. Temperature is the only free parameter — the functional form is completely determined by the five axioms.
- **Lecture 7:** The SFM architecture prescribes a learned temperature following CLIP. Temperature recovery is introduced as a testable prediction connecting the learned model to the physical system.
- **Lecture 10:** The affinity calibration framework uses the learned τ to derive the temperature recovery prediction: $w_B / w_A = \tau_{\text{learned}}$. The physical interpretation of τ is tested quantitatively.